LLM reads, code verifies

A routing and verification harness for receipt OCR: when a cheap local reader is enough, when to escalate, and how much model error deterministic code catches

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Portfolio & resume

Code & results

1 PROBLEM

I built the receipt-OCR side of a production expense platform (20 employees, about 200 receipts a month) that cut reconciliation time by 90%+, as the sole engineer over 729 merged pull requests.

A vision LLM reads receipts well but grades its own work badly: its self-reported confidence is not evidence, and an auto-approval path built on it was rolled back. **When is a cheap local reader enough, when should a receipt escalate, and how much error can code catch with no answer key?**

2 FROM PRODUCTION

- Built a 90-receipt, 8-currency golden set by hand; benchmarked 7 LLMs on accuracy, stability, cost and latency.
- Gemini-to-Claude fallback with a fast-fail budget: a 503 storm that pushed one receipt to 87 s now bounds at about 10 s.
- +11% amount-extraction pass rate from fixing a validator that silently dropped 12 model-returned fields.

3 EXPERIMENT

- **300 scanned receipts** (ICDAR 2019 SROIE) and **200 phone photos** (WildReceipt). Exact match on date and total; scoring is code, never a model.
- **43 synthetic adversarial receipts**: future and ancient dates, arithmetic traps, prompt injection in English and Korean. Held out; gates are set on the natural data only.

Terms. Oracle router: a router that knows the answers and escalates exactly the receipts the local reader gets wrong; the ceiling any real signal can reach. **Oracle gap**: distance between that ceiling and the real router. **Accepted precision**: share of auto-accepted answers with date and total both right. **Leak**: a wrong answer the code checks accepted. **AUC**: area under the ROC curve of the learned score that predicts "the local reader is wrong", out of fold.

4 A LADDER OF READERS, A WALL OF CHECKS

T1 · local OCR + rules
RapidOCR / Tesseract on CPU. $0, 5 s.

T2 · local VLM
Qwen2.5-VL 3B on a 4 GB GPU. $0, 28 s.

T3 · cloud, cheap
Gemini 3.1 Flash-Lite. $0.54 per 1k.

T4 · cloud, strong
Gemini 3.6 Flash, thinking off. $1.42 per 1k. Vendor failover to Claude.

Code checks on every model answer (no answer key, no model confidence):

- sanitizer: injection patterns in EN/KO/JA/ZH, field whitelist, types
- arithmetic: subtotal + tax + tip + discount = total; absent is not zero
- date plausibility
- two-reader agreement: the model's total and date must appear in the OCR text

Zero flags → **auto-accept**. Flagged → escalate one rung → still flagged → **human review**.

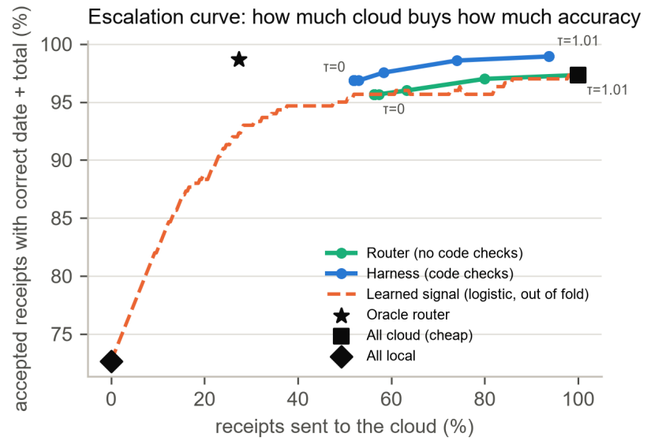


Fig A. Receipts sent to the cloud (x) vs accepted precision (y) as the local confidence threshold sweeps. The star is the oracle; the gap between it and the curves is what better escalation signals could still buy.

ROUTING POLICY (300 SCANS)	CLOUD	REVIEW	ACCEPTED RIGHT	USD / 1K
Oracle (knows the answers)	27%	0%	98.7%	0.15
Rule router: both OCR readers agree, confidence ≥ 0.9	56%	0%	95.7%	0.30
Rule router + code checks (harness)	52%	4%	96.9%	0.37
Learned signal, 13 features, AUC 0.89 (out of fold)	61%	0%	95.7%	0.33
Local only: T1, then local VLM with checks, else review	0%	19%	93.4%	0

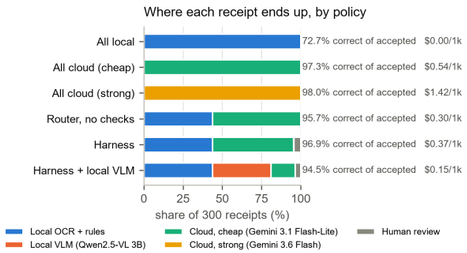


Fig B. Where 300 scanned receipts end up under each policy, the share of accepted answers that are right, and API cost per 1,000.

5 RESULTS

READER	SCANS: DATE + TOTAL	PHOTOS	MEDIAN
Local OCR + rules	72.6%	55.0%	4.7 s
Local VLM 3B	82.9%	81.0%	28.3 s
Cloud, cheap	97.3%	86.5%	1.8 s
Cloud, strong	98.0%	86.5%	1.9 s

ADVERSARIAL SET (CLOUD, CHEAP)	N	MODEL FOOLED	GATE FIRED	LEAKED
Date traps	10	0	10	0
Arithmetic traps	11	1	11	0
Injection, blatant	6	0	6	0
Injection, disguised	10	0	10	0
Controls	6	0	6	0

6 FINDINGS

- **Cheap readers get far, not far enough.** Local OCR + rules reads date and total right 73% of the time at $0, a local 3B vision model 83%, cloud 97 to 98%. On phone photos every reader drops: local 55%, cloud 87%.
- **The escalation signal, not the model, is the bottleneck.** A perfect router would send 27% to the cloud for 98.7%; the rule router sends 56% for 95.7%, and a learned score with AUC 0.89 does no better. The strong cloud tier adds 0.7 points for 2.6x the price: escalate to people, not to a bigger model.
- **Code catches what the model cannot see about itself.** With no answer key the checks caught 38 of the local model's 51 wrong answers and 5 of the cloud's 8. Of 18 that slipped through, 16 were a different printed amount, not a misread. All 16 injections were ignored by every model; every date and arithmetic trap tripped a gate; nothing leaked.

Next. Other currencies, date conventions and document types (invoices, statements) move every number here, so the gates and the sample are re-tuned per domain; 10k receipts; layout features as escalation signals; the oracle gap as a weekly drift metric.

Sources. ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), CC-BY-4.0 · WildReceipt (Sun et al., arXiv:2103.14470) · Tesseract 5.4, RapidOCR / PP-OCR, Qwen2.5-VL (Apache 2.0) via Ollama · Google Gemini API pricing page, 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda (MIT). Production figures are from the author's 2026 engagement; the platform is private.

Run of 2026-10-31 · code, manifests and every result are in the repository